

(a) Final code and repair outcome

Autoencoder: Linear $\rightarrow$ Tanh $\rightarrow$ Linear

Loss difference: 1.43×10^{-6} before and after repair

SWE-agent / MatchFixAgent

Tanh mapping still absent; existing ReLU mapping:

```
@register_function(torch.relu)
@register_function(torch.nn.functional.relu)
def functional_relu(input, inplace=False):
    return mops.relu(input)
```

LaDiM

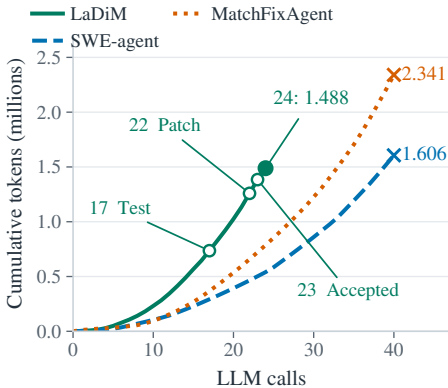
Added Tanh mapping; accepted on all three seeds

```
@register_function(torch.tanh)
@register_function(torch.nn.functional.tanh)
def functional_tanh(input):
    return mops.tanh(input)
```

Gradient vector difference: $0.359 \rightarrow 1.01 \times 10^{-6}$

Relative parameter update difference: $0.771 \rightarrow 6.42 \times 10^{-6}$

(b) Repair process and cost



Accepted: public test passes

24 calls: final report completed

× Unrepaired at the 40-call limit